

Linear Circuit Analysis**(EE1001)**

Date: February 29, 2024

Course Instructor(s)

- | | |
|------------------------|------------------|
| 1. Dr. Saima Zafar | 2. Mohsin Yousuf |
| 3. Beenish Fatima (CM) | 4. Akbare Yaqub |

Sessional-I Exam

Total Time: 1 Hour

Total Marks: 40

Total Questions: 02

Semester: SP-2024

Campus: Lahore

Dept: Electrical Engineering

Student Name

Roll No

Section

Student Signature

Vetted by

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Instruction/Notes:

1. Show all steps with proper circuit diagrams.
2. Attempt all parts of a question together.

CLO # 01: Analyze simple resistive circuits using basic network laws**(C4)**

Q1: For the circuit shown in the Figure 1 below, illustrate the use of voltage & current division rules (VDR and CDR) along with power computation. **[20 marks]**

(a) Find i_1 and i_4 by CDR.**[10]**(b) Find v_2 by VDR.**[6]**

(c) Find the power delivered by the current source.

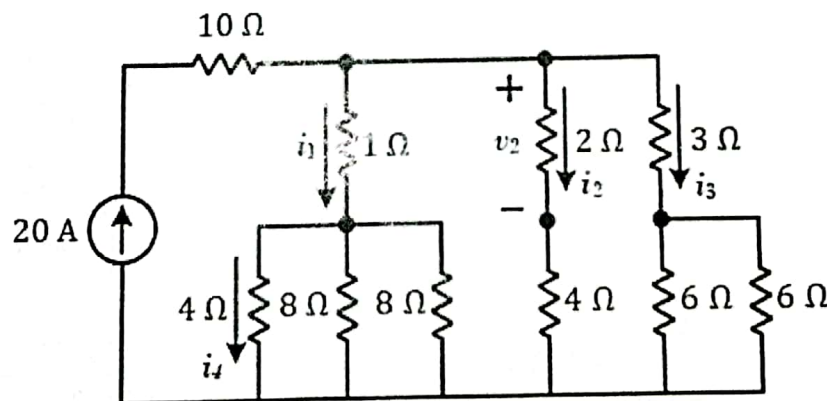
[4]

Figure 1. Circuit with Resistors

$i_1 = 1.7$
 $i_2 = 0.88$
 $v_2 = 1.76$
 $i_4 = 0.86 A$

CLO # 01: Analyze simple resistive circuits using basic network laws

(C4)

Q2: For the circuit shown in the Figure 2 below, appraise the use of Ohm's Law and Kirchhoff's Law (KVL and KCL) along with power computation. [20 marks]

- (a) Find i_1 and i_2 . [10]
(b) Find v_1 and v_2 . [4]
(c) Name the dependent source and compute the power associated with both the independent and dependent source. [6]

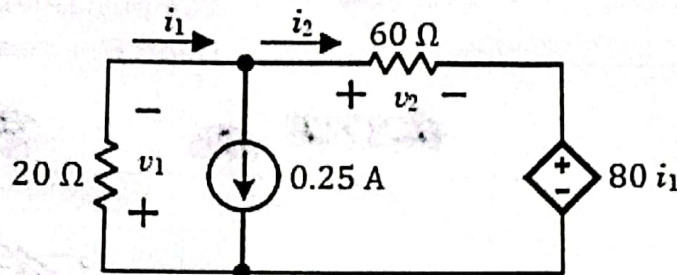


Figure 2. Circuit containing Dependent Source

~~$i_1 = 0.01 \text{ A}$~~
 ~~$i_2 = 0.01 \text{ A}$~~
 ~~$v_1 = 5 \text{ V}$~~
 ~~$v_2 = 10 \text{ V}$~~